A guide to designing germline-dependent epigenetic inheritance experiments in mammals

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Recent work has demonstrated that environmental factors experienced by parents can affect their offspring across multiple generations, and that such transgenerational transmission can depend on the germline. Causal evidence for the involvement of germ cells is rare, however, and the underlying molecular mechanisms remain poorly understood. Further, studies often employ varying methods in experimental design and data interpretation. We provide a critical analysis of these issues and suggest possible solutions and guidelines for improving study design and generating reproducible and high-quality data.

During sexual reproduction, genetic material is transferred from parent to offspring. In addition, mechanisms independent of the DNA sequence—but still dependent on the germline—also contribute to transmission of acquired features from parents to offspring[1]. Over the last 5 years, such germline epigenetic inheritance has been reported in different species, including worms, rodents and humans (for recent reviews, see refs. 1–4). However, the concept of transmission of acquired traits is still facing criticism[5,6], mostly because studies often lack causal evidence that epigenetic processes that control germ cells are responsible for the transmission and expression of acquired traits in the offspring. Further, because the field is highly interdisciplinary and usually combines domains like epigenetics, physiology, neuroscience, immunology, reproductive biology and/or endocrinology; vastly different models, techniques and methodological approaches are used. This makes it challenging to ensure similarly high standards and to generalize conclusions across disciplines. Here, we address these issues and discuss key problems in the design and conduct of studies in transgenerational epigenetic inheritance in rodents. We formulate major questions that often arise in the field and provide guidelines for future work.

Definitions and terminology

'Germline epigenetic inheritance' (i) involves epigenetic mechanisms and (ii) depends on germ cells. (i) Because the definition of epigenetics is often debated[7,8], this form of inheritance is also called 'germline nongenetic inheritance'[1,9]. Here, epigenetics is defined as the processes and marks on or around the DNA processes that control the activity of the genome and can be mitotically and/or meiotically inherited. Genome activity not only includes gene expression but also the regulation of transposable elements and of repeat sequences in heterochromatin and the control of chromatin structure. The mechanisms underlying germline epigenetic inheritance include DNA methylation[10–15], noncoding RNAs (ncRNAs)[16], histone (perhaps also protamine) post-translational modifications and histone variants[17,18], and possibly other yet unknown epigenetic processes (reviewed in refs. 1, 19 and 20). We refer to alterations in these marks as epimodifications, and they can be dependent or independent of genetic mutations[21,22]. In addition, other factors like maternal proteins in oocytes, prion proteins or microbiota may also contribute to germline epigenetic inheritance, but these possibilities remain speculative[23,24]. (ii) The emphasis on the involvement of the germline is critical to focus on mechanisms that

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engage biological inheritance[25]. This is in contrast to the transfer of characters or features to the offspring through social or cultural means[26] that do not involve the germline. A classic example of germline-independent epigenetic inheritance in rodents is the induction of epigenetic changes in the offspring by variations in maternal care[27]. Conversely, a classic example of germline epigenetic inheritance is imprinting, in which epigenetic information transferred by the germline leads to monoallelic gene expression in a parental origin-specific manner[28]. Proper experimental design is critical for probing the germline dependence of environmentally induced changes (**Supplementary Note**).

Experimental design

Transgenerational study designs in mammals are notoriously difficult because they require long and complex breeding schemes and careful planning. Numerous pitfalls that can interfere with experimental results and limit their interpretation also have to be considered. Studies on germline epigenetic inheritance must include a particularly rigorous assessment of potential alternative modes of transmission that are germline independent but could be interpreted as germline dependent.

Matriline versus patriline. Environmentally induced changes may be inherited from mother, father or both parents. Breeding designs that assess these options require exposing females and males of the parent generation to the environmental factor under study (e.g., experimental diet and control diet), then breeding the exposed females to control naive males (matriline) and exposed males to control naive females (patriline). A 'dual breeding group' in which exposed males are bred with exposed females can also be included to test for possible interacting effects that may be different than effects resulting from transmission through only one parent[29], but dual breeding should always be complemented by matrilineal and/or patrilineal breeding. In all cases, a parallel control group is always required where nonexposed parents are mated. Such inclusive designs using four groups are recommended when first investigating transmission of an acquired trait and have revealed cases of transmission through only the dam or the sire[30,31] or through both exposed parents[29]. It is important to consider, however, that this design will generate large numbers of animals. If 10 breeding pairs are used per group, this generates 40 breeding pairs in total and up to 400 pups in the first (F1) generation (256 pups for a pregnancy rate of 80% and mean litters of 8 pups). This number doubles when grandoffspring (F2) are generated. It is therefore highly demanding in terms of workload, cost and animal space. If transmission occurs through both females and males, patrilines are often chosen because of experimental advantages (see below) and easier analyses given the abundance of male germ cells. The advent of single-cell epigenetics and transcriptomics nonetheless greatly facilitates the analyses of oocytes, and transmission through females should actually receive equal attention in research.

Duration of cohabitation during breeding. In matrilineal studies, it is key to determine whether effects are mediated by the germline or confounded by prenatal and/or postnatal parameters such as the intrauterine milieu and/or maternal behaviors. Such germline-independent parameters can have profound, lifelong consequences for the offspring[32–34]. To avoid these, many researchers use patrilines; as in mice and rats, males contribute minimally to rearing offspring and can be separated from the female after mating, thus avoiding interaction with the offspring. But treatments like chronic exposure to endocrine disruptors[35], stress–trauma paradigms[10,30,36,37], drugs of abuse[17,38,39], infections[18] or dietary challenges[14,15,40–42] used in epigenetic inheritance studies can alter appearance, health and behavior of males. This may affect females during mating and have consequences for the offspring, since female rodents can modify their maternal investment depending on qualities of their mate[32,43,44]. For example, unusual social interactions with a male may activate the release of stress hormones or cytokines in the female and ultimately impact developing embryos *in utero* and/or maternal behaviors after delivery. Such effects in the offspring may mistakenly appear to be transmitted by the father[45]. To minimize the potential impact of the male on the female, the male can be removed when a vaginal plug (indicating successful mating) is detected[37,39]. However, plug checking involves handling and implies different cohabitation time between breeding pairs, which introduces biases and is usually not reported in studies using such breeding schemes. Additionally, short cohabitation time can lower breeding success, and may in turn necessitate larger breeding cohorts to obtain enough pups. Alternatively, dam and sire can also be maintained together throughout gestation and even after birth[30], which may provide a social advantage but introduces biases due to interactions between male and offspring (particularly in patrilineal studies). A compromise is to keep breeders together for 4–5 d, allowing the females to undergo at least one estrus cycle to maximize breeding success[10,11]. To most effectively reduce cohabitation time while ensuring breeding success, we recommend to pair male and female only after inducing natural estrus by exposing females to male pheromones present in urine[46–49]. Then females can be paired with males for breeding for 12–24 h, resulting in reasonably high pregnancy rates[48]. Nonetheless, sires could still impact the dam during short periods of cohabitation. Definitively ruling out any such effects requires the use of assisted reproductive techniques (ARTs). Finally, a potential but rarely considered confounder is the time the gestating dam spends alone in the cage after the sire is removed. Prolonged single housing in rodents is a potent stressor in itself[50,51]. This can be avoided by using breeding trios with one male and two females, so that dams remain in pair housing after the male is removed. After delivery, individual dams and pups can be transferred to new cages, allowing for cross-fostering when needed.

Causal proof of germline-dependent transmission. Different experimental strategies are necessary to test the various factors that can impact inheritance independently of the germline. Controlling the impact of the mother on the offspring is essential for both matrilineal and patrilineal designs. Postnatal maternal effects can be revealed by cross-fostering, a procedure that should be employed not only for matrilineal but also for patrilineal studies to control the impact of the male on postnatal maternal investment[11,52]. However, maternal effects can be exerted on the offspring *in utero*, for example via changes in circulating hormones, lipids or other blood-borne factors. Therefore, proof that a heritable effect is indeed germline dependent requires the use of ARTs (**Fig. 1**).

Patriline. ARTs such as artificial insemination and *in vitro* fertilization (IVF) are extremely useful for patrilineal studies because they eliminate interactions between male and female (**Fig. 1a**).

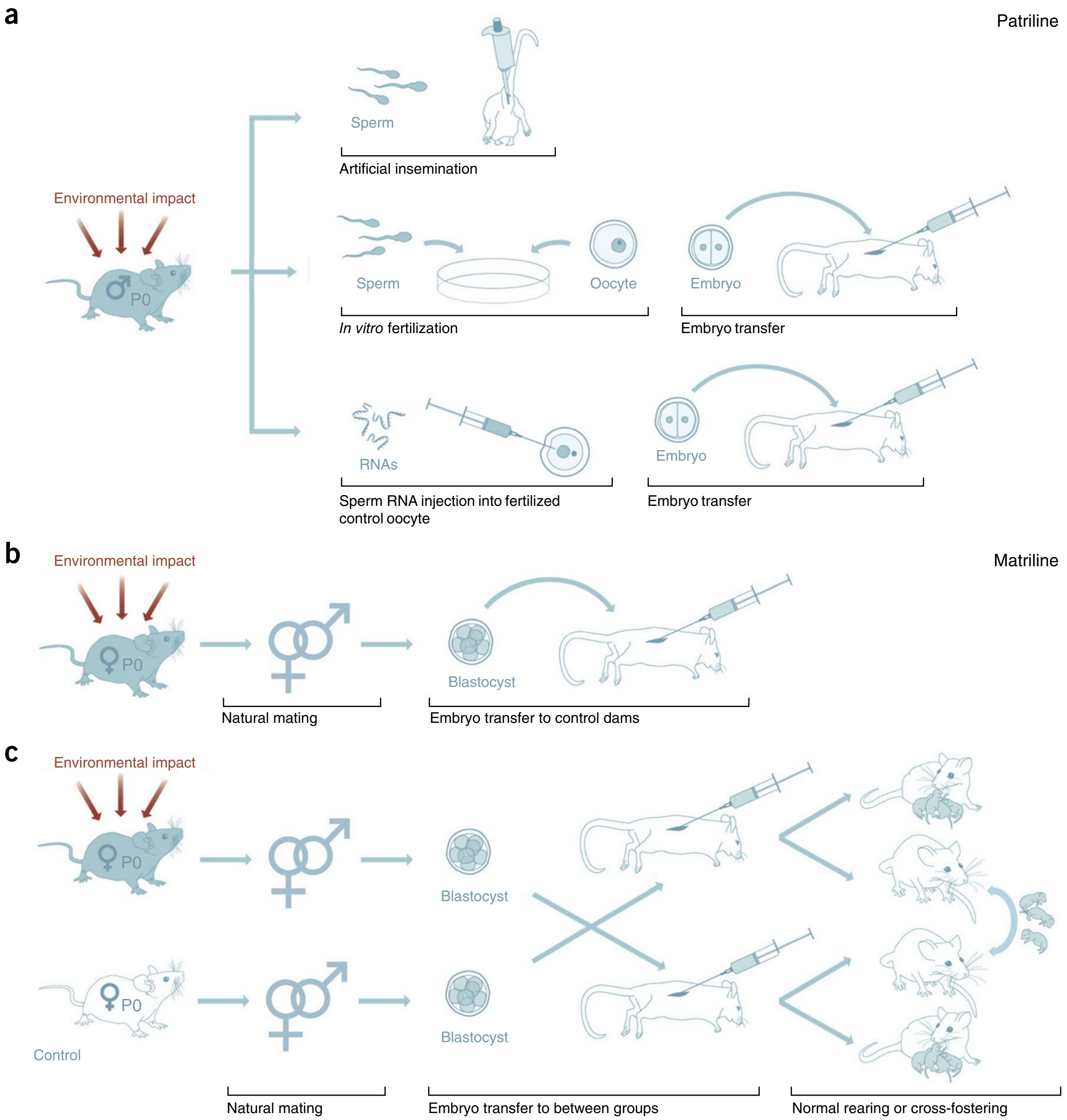

Figure 1 | Experimental strategies to causally test germline dependence of inherited phenotypes. (**a**) Three approaches for patrilineal designs to test whether transmission occurs through the germline. (**b**) Embryo transfer in matrilineal designs. (**c**) Embryo transfer between groups and cross-fostering to look at the contribution of intrauterine or postnatal effects. (Copyright by S. Steinbacher, University of Zurich, Information Technology, MELS/SIVIC.)

If an environmentally induced effect is transmitted to the offspring via artificial insemination or IVF, this is causal proof that it involves the germline[36,48,53,54]. Importantly, working with patrilines also allows extracting certain factors from sperm that can be injected into oocytes fertilized by naive (control) males[55,56]. Using this strategy, sperm RNAs were revealed to be nongenetic vectors that allow the transfer of acquired information from father to offspring in a purely germline-dependent manner[16]. However, one major caveat is that ARTs can induce widespread epigenetic alterations. Procedures like ovarian stimulation for superovulation, *in vitro* maturation of oocytes, cryopreservation and *in vitro* embryo culture used for ARTs can alter the epigenetic profile of germ cells or embryos[57]. IVF uses several of these procedures and thus carries a high risk for epigenomic interference. This is a serious confounding factor, preventing any valid conclusion about germline dependence if a transgenerational effect is not reproduced using IVF[1,36]. One solution to this is the use of artificial insemination, which avoids confounding factors like *in vitro* culture and can be conducted without surgery[58]. A recently refined protocol in mice allows artificial insemination without superovulation or breeding to vasectomized males, further minimizing confounding factors and greatly simplifying study design[48]. This protocol, which requires no expensive equipment or specialized reproductive facilities, is strongly recommended to avoid inherent biases of ARTs and to efficiently test germline dependence in epigenetic inheritance models.

Matriline. While it is easy to eliminate interactions between females and males during mating in patrilineal studies, matrilineal designs are more demanding because they require controlling for *in utero* and postnatal effects. One way to eliminate these effects is to use embryo transfer[59–62]. This involves surgically extracting embryos from a gestating female, followed by transfer (implantation) into a control naive foster mother (**Fig. 1b**). An unexposed control group undergoing embryo transfer is always required. An issue with embryo transfer, however, is that it uses superovulation and *in vitro* culture, known to alter the epigenome[57]. This may have consequences for the resulting embryos, a fact that is often ignored but should be considered. Surgical implantation may also introduce confounds, but these can be avoided by using novel nonsurgical techniques of embryo transfer[63,64] combined with natural induction of ovulation[48]. Failure to reproduce a transgenerational effect after embryo transfer may indicate germline independence or epigenetic confound interference (which in some cases may also lead to unexpected phenotypes). In this case, it is necessary to determine whether the original transgenerational effect depends on *in utero* effects or postnatal maternal care. This can be done by using dams from the experimental as well as control group as both donors and recipients during embryo transfer and then cross-fostering some of the pups back to their biological mothers (**Fig. 1c**). Such designs have previously revealed clear dissociation between transmitted effects dependent on *in utero* exposure and postnatal care[65].

Selecting animals for breeding. We will illustrate this rarely discussed issue with a simple example. An experimenter wants to investigate the effects of chronic stress experienced by adult male mice on the offspring's level of anxiety. In the parent generation (P0), 20 male mice are exposed to chronic stress; 20 males are controls. As for most environmental factors, stress does not affect all individuals the same way, but it can have variable effects even on inbred mice[66,67]. Assume that 10 out of 20 stress-exposed males are mildly anxious in adulthood; 5 mice show severe anxiety, and 5 mice are not anxious at all. There are now at least four options for breeding. (i) All 20 mice from each group are used for breeding, resulting in a large cohort of offspring and possibly higher variance caused by variability between fathers. This strategy is demanding in terms of colony size but may reveal interesting insights about, for instance, resilient animals. (ii) Males are selected based on behavioral extremes in both groups, such that the 10 most anxious males from the stressed group and the 10 least anxious control mice are used for breeding. This may help isolate an effect that may be shadowed by natural variations in paternal behavior transmitted to the offspring, even in inbred mice[68]. However, because genetic and epigenetic differences also exist in inbred mice[22], selecting breeders based on behavioral extremes increases the risk of accentuating undetected genetic differences and biases the interpretation of results. (iii) Breeders can be selected randomly from each group or based on anxiety levels closest to the group mean. However, with all these approaches it is difficult to choose the right behavioral parameter for selecting animals, since behavioral responses of a given animal can vary greatly between different tests. Thus, an animal with high anxiety on one test may show moderate anxiety on another test. Thus, behavioral batteries should be used when selecting animals based on behavioral performance, and animals with consistent behaviors on all tasks should be selected if possible. Another issue is that behavioral testing might itself introduce confounds that could interfere with the transmission of certain traits over generations. Therefore, mice should be allowed to recover from behavioral testing for several weeks before mating. Highly stressful behavioral tests involving foot shock or painful procedures should be avoided before breeding. (iv) To eliminate the influence of behavioral testing, behaviorally naive breeders can be used[15,31,54]. Thus, different animals from a given litter are selected for behavioral tests, molecular analyses or breeding. We favor this approach because it prevents the altering of transgenerational effects by possible confounds arising from prior behavioral testing. The drawback is that no behavioral or physiological data are available from breeders that constitute the parent generation (P0), so it requires that the phenotype(s) be very consistent and robust to guarantee that breeders do express the desired phenotype, even if not tested.

Outbred versus inbred strains. Pioneering work demonstrated that outbred rats and mice are more sensitive to the transgenerational impact of the endocrine disruptor vinclozolin than an inbred strain[35,69]. Prominent examples of diet-induced transgenerational effects in mice use the outbred CD1(ICR) strain[14,70]. Early work showed that the axin-fused ($Axin^{Fu}$) kinked-tail phenotype can be inherited through the patriline only when males are bred to 129P4/RrRk dams and not when they are bred to C57Bl/6J dams[71]. This suggests that germline epigenetic programming varies between strains, although this hypothesis has not been tested beyond the $Axin^{Fu}$ locus. Some studies have since—in response to this work—opted to breed C57Bl/6J males to 129Sv/ImJ dams[38], or have used C57Bl/6:129 F1 hybrids[72]. However, using hybrids means that father, offspring and grandoffspring are genetically different, complicating transgenerational epigenetic analyses and the interpretation of behavioral results. Many studies of germline epigenetic inheritance instead use inbred C57Bl/6 mice in both sexes[10,11,13,16,36]. This strain allows genetic manipulations for testing the contribution of specific genes or of the interaction of genes and environment to epigenetic inheritance. Indeed, some transgenic models show germline epigenetic inheritance of specific symptoms in the wild-type offspring and grandoffspring of transgenic ancestors[61,73]. Because the original mutation is known in these models, they may prove particularly useful for uncovering novel mechanisms underlying transmission. Therefore, unless there are clear reasons to choose an outbred strain, we recommend using inbred strains to remove genetic variability and aid the interpretation of epigenetic data.

Avoiding litter and cage effects. In general, mice and rats from a given cage or from the same litter tend to be more similar (behaviorally and molecularly) than those from other cages and litters[74,75]. To avoid confounding litter and cage effects in the offspring, careful experimental design is needed at the time of weaning, when male and female pups are separated from the mother, split by gender and assigned to new cages. A key decision at weaning is whether same-sex siblings of a given litter remain together (**Fig. 2**, left side) or are split and mixed with same-sex conspecifics from other litters of the same experimental group (**Fig. 2**, right side). Each strategy has advantages and drawbacks. Weaning pups from the same litter into the same cage (**Fig. 2**, left side) means that cage and litter effects are indistinguishable. Thus, biased results due to cage and litter effects can be avoided simply by assigning each animal from each cage to a different test (behavioral or molecular phenotyping, breeding, etc.; see **Fig. 2**, bottom row)[31]. Then, each

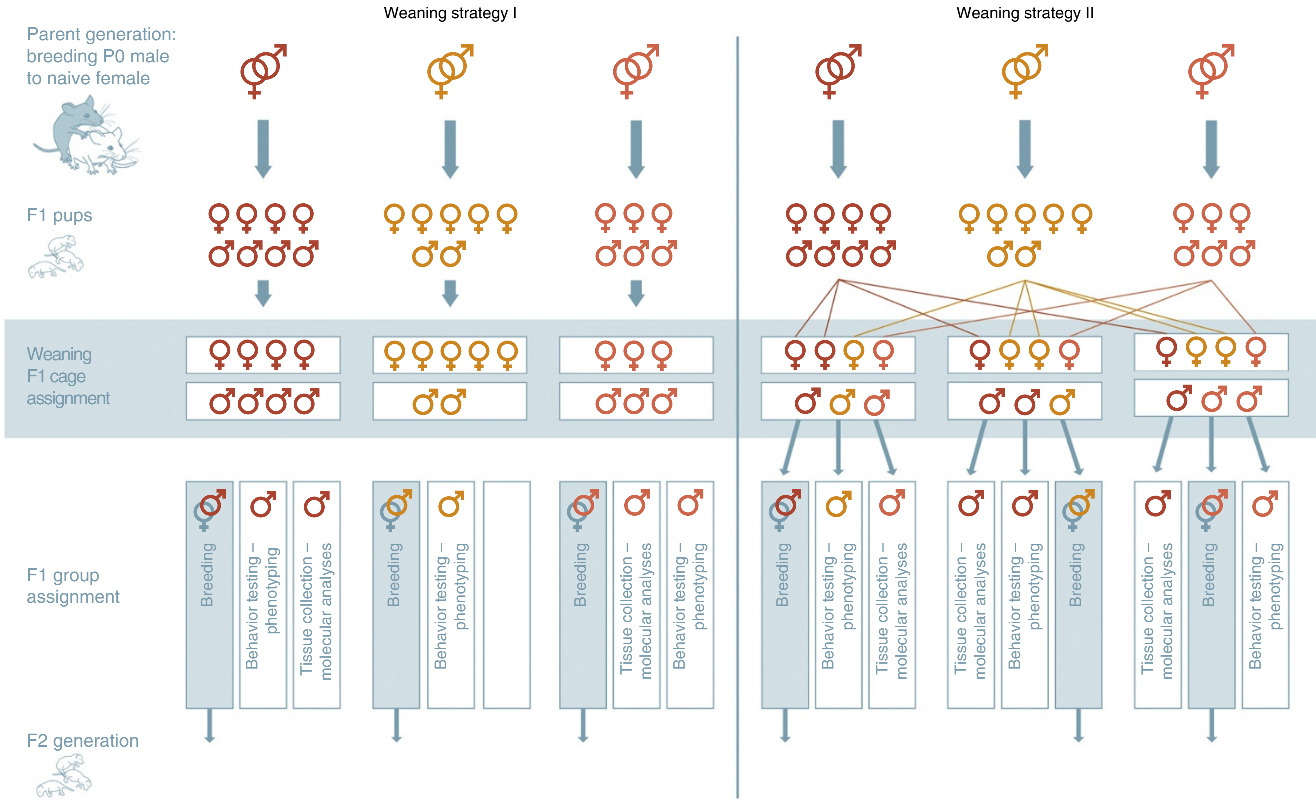

Figure 2 | Weaning strategies for rodent studies of germline epigenetic inheritance. Same-sex siblings of a given litter are kept together (Weaning strategy I) or mixed with same-sex conspecifics from other litters (Weaning strategy II). On each side, three breeding pairs from the same experimental group are depicted; different colors indicate different fathers and allow tracing the offspring back to their parent of origin. Assigning different mice from a given litter to different tests (bottom row) ensures that the dam is correctly used as the experimental unit in subsequent analyses. (Copyright by S. Steinbacher, University of Zurich, Information Technology, MELS/SIVIC.)

animal in each test automatically comes from a different litter and a different cage. However, this weaning strategy inevitably results in different numbers of animals per cage, as rodents tend to have litters of different size and variable male–female ratios (**Fig. 2**, "Weaning" row). The number of animals per cage, however, is an important factor that impacts behavior[76] and should be kept constant within and between experimental groups. Additionally, to avoid group effects, it is recommended to use mice from different cages for all tests; but in a cage with only two animals options are limited (**Fig. 2**, left side). To solve this issue, pups can be culled after birth or before weaning to generate balanced numbers, but this reduces the amount of animals available for later testing. Alternatively, it is possible to mix pups from different litters upon weaning to have equal numbers of animals per cage (**Fig. 2**, right side). This strategy avoids culling and maintains similar numbers of animals per cage, yet care must be taken to assign offspring from different litters for subsequent testing to avoid litter effects. Independent of the weaning strategy, it is also necessary to label animals to allow assignment to different tests or analyses and to trace the litter of origin. Details about weaning strategy should be reported in published work.

Group size. To determine group size (or number of experimental units) for statistical analyses, it is necessary to estimate the expected size and variability of the effect(s) based on pilot studies, published data or expert opinion[77]. The experimental unit is defined as the smallest division of the experimental material such that any two experimental units can receive different treatments. Therefore, in a patrilineal design, the individual male mice constitute the experimental unit (n). For generating offspring, a female is randomly assigned to either exposed or control sires, but her pups are all assigned to the same group; therefore, n is the dam. Since they are assigned to the same group, all pups of a given litter must statistically be considered one sample. Considering pups (and not dams), the experimental unit will lead to an inflated type I (alpha) error rate[74,78]. This applies not only to the offspring, grandoffspring and all subsequent generations of an exposed individual but also to the exposed generation itself if exposure involves manipulations of gestating dams or newly born pups[78]. In the case of *in utero* exposure (e.g., stress during gestation), n is the exposed mother, despite the fact that her pups are also directly exposed and used to generate offspring and grandoffspring. This issue needs to be addressed during experimental planning by using power analyses to estimate the number of litters required for the experiment, counting the number of litters in each generation as the experimental unit. Then, animals of different litters can be assigned to different behavioral or molecular tests and for breeding[31] (**Fig. 2**). Alternatively, multiple animals from the same litter can be tested, but their scores should either be averaged and considered as a single sample, or researchers can employ a mixed-effects model of analysis, which ensures that animals are nested within litters[74].

Tissues to study in parent and offspring

It is challenging to select specific tissues for molecular analyses in parent and offspring, because inherited epimodifications can potentially manifest in any tissue, and the mechanisms guiding tissue specificity are unknown. Further, it remains unclear in which reproductive cells epimodifications can be installed—particularly in males, where continuously renewing germ cells go through several developmental stages[1]. We review the anatomy and timing of spermatogenesis (**Supplementary Note 1** and **Supplementary Fig. 1**) and explain how it should inform study design and the type of cells or tissue analyzed in parents. We also provide suggestions to guide tissue collection for molecular analysis in the offspring.

Outlook and conclusions

Germline-dependent epigenetic inheritance is an extremely dynamic and exciting field that is expected to shift the fundamental understanding of inheritance and provide novel insight into the etiology, expression and risk of many complex diseases[1,79]. Future research will need to address three major challenges. It must: (i) identify specific germline epimodifications causally responsible for the transfer of information to the offspring. To gain such causal evidence, recent technological advances in (epi)genome editing offer the outstanding possibility of manipulating epimodifications and testing their functional involvement in the transmission of traits across generations[80,81]. (ii) Identify the mechanisms by which environmental factors can install and maintain epimodifications in germ cells. This will require studying multiple tissues and testing different possible links between these tissues and developing germ cells. (iii) Determine the contribution of genetic factors to epigenetic inheritance by delineating the interaction between genetic mutations and epimodifications. This will require high-throughput genomic methodologies to screen and identify genetic features and to integrate complex genomic and epigenomic data sets[82]. Recent work has revealed the potential of this approach by demonstrating that diet-induced changes in DNA methylation can depend on a single nucleotide variant in ribosomal DNA[21]. Innovations in the field will also need to use rigorous standards for experimental design and data interpretation. This will be critical for gaining new and accurate knowledge about germline dependence and the mechanisms underlying transgenerational epigenetic inheritance and for providing a solid basis for the paradigm shift in biology that this research offers.

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The authors declare no competing financial interests.

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